ENZYMES

Definition, Classification, Active Site, Cofactors, Proenzymes and Mechanism of Action

High-yield definition

Enzymes are biocatalysts synthesized by living cells. They increase the rate of biochemical reactions without being consumed, are usually protein in nature, are thermolabile, colloidal and highly specific in action. Exception: some RNA molecules act as ribozymes.

1. Definition and General Characters

- **Biocatalyst:** a catalyst of biological origin that speeds up chemical reactions in living systems without undergoing permanent change.
- **Protein nature:** most enzymes are proteins; their tertiary structure creates the correct catalytic shape.
- **Specific action:** each enzyme acts on a particular substrate or a group of related substrates.
- **Thermolabile:** enzyme activity is lost at high temperature due to denaturation.
- **Reusable:** after product formation, the enzyme is released and can catalyse another reaction.
- **Large molecule, small active centre:** only a small region of the enzyme participates directly in substrate binding and catalysis.

2. Classification of Enzymes with Examples

The International Union of Biochemistry classifies enzymes into six major classes according to the type of reaction catalysed. A useful mnemonic is OTHLIL: Oxidoreductases, Transferases, Hydrolases, Lyases, Isomerases and Ligases.

Class	Main reaction	Examples	Simple reaction form
1. Oxidoreductases	Catalyse oxidation-reduction reactions	Alcohol dehydrogenase, cytochrome oxidase, amino acid oxidases	AH2 + B -> A + BH2
2. Transferases	Transfer functional groups from one molecule to another	Hexokinase, transaminases, transmethylases, phosphorylase	A-X + B -> A + B-X
3. Hydrolases	Break bonds by addition of water; hydrolysis reactions	Lipase, pepsin, urease, phosphatases, cholinesterase	A-B + H2O -> AH + BOH
4. Lyases	Add or remove groups such as H2O, NH3 or CO2 without hydrolysis	Aldolase, fumarase, histidase	A-B + X-Y -> AX + BY
5. Isomerases	Catalyse interconversion of isomers	Triose phosphate isomerase, phosphohexose isomerase	A -> A'
6. Ligases	Join two molecules using ATP; synthetic reactions	Glutamine synthetase, acetyl-CoA carboxylase, succinate thiokinase	A + B + ATP -> A-B + ADP + Pi

The source textbook table for classification is shown below for visual revision.

Enzyme class with examples*	Reaction catalysed
1. **Oxidoreductases**	
Alcohol dehydrogenase (alcohol : NAD+ oxidoreductase E.C. 1.1.1.1.), cytochrome oxidase, L- and D-amino acid oxidases	Oxidation → Reduction $AH_2 + B \rightarrow A + BH_2$
2. **Transferases**	
Hexokinase (ATP : D-hexose 6-phosphotransferase, E.C. 2.7.1.1.), transaminases, transmethylases, phosphorylase	Group transfer $A - X + B \rightarrow A + B - X$
3. **Hydrolases**	
Lipase (triacylglycerol acyl hydrolase E.C. 3.1.1.3), choline esterase, acid and alkaline phosphatases, pepsin, urease	Hydrolysis $A - B + H_2O \rightarrow AH + BOH$
4. **Lyases**	
Aldolase (ketose 1-phosphate aldehyde lyase, E.C. 4.1.2.7), fumarase, histidase	Addition → Elimination $A - B + X - Y \rightarrow AX - BY$
5. **Isomerases**	
Triose phosphate isomerase (D-glyceraldehyde 3-phosphate ketoisomerase, E.C. 5.3.1.1), retinol isomerase, phosphohexose isomerase	Interconversion of isomers $A \rightarrow A'$
6. **Ligases**	
Glutamine synthetase (L-glutamate ammonia ligase, E.C. 6.3.1.2), acetyl CoA carboxylase, succinate thiokinase	Condensation (usually dependent on ATP) Condensation (usually dependent on ATP) $A + B \xrightarrow[\text{ATP} \rightarrow \text{ADP + Pi}]{} A - B$

Source Table 6.1: Classification of enzymes with examples and reaction types.

3. Active Site

Definition

The active site or active centre is the small region of an enzyme where substrate binds and catalysis occurs. It usually contains a substrate-binding site and a catalytic site.

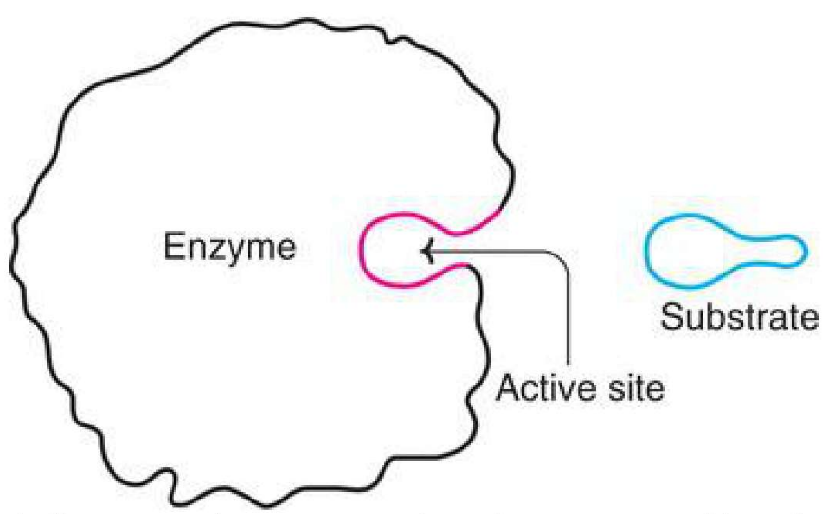

FIG. 6.6 A diagrammatic representation of an enzyme with active site.

Source Fig. 6.6: Enzyme with active site and substrate.

Salient features of active site

1. The active site is produced by the three-dimensional tertiary structure of the enzyme protein.
2. It is usually a cleft, crevice or pocket in the enzyme molecule.
3. It contains amino acid residues called catalytic residues; these may be far apart in the primary structure but come together after protein folding.
4. It is not rigid; flexibility helps specific substrate binding.
5. Substrate binds mainly by weak non-covalent bonds such as hydrogen bonds, ionic interactions and hydrophobic interactions.

6. Cofactors or coenzymes required by some enzymes may be present as part of the catalytic site.
7. Common active-site amino acids include serine, histidine, cysteine, lysine, arginine, glutamate, aspartate and tyrosine.
8. Active sites explain enzyme specificity: only suitable substrates fit and react efficiently.

4. Cofactors, Coenzymes, Apoenzyme and Holoenzyme

Many enzymes require a non-protein component for full catalytic activity. These additional components are collectively called cofactors.

Term	Meaning	Examples
Apoenzyme	Protein part of an enzyme; inactive alone if a cofactor is required.	Protein portion of dehydrogenase enzyme
Cofactor	Non-protein component required for enzyme activity; may be inorganic or organic.	Mg^{2+}, Zn^{2+}, Fe^{2+}, NAD^+, FAD
Coenzyme	Organic, low molecular weight, dialysable cofactor; often derived from B-complex vitamins.	NAD^+, $NADP^+$, FAD, FMN, TPP, CoA
Prosthetic group	Non-protein group tightly bound to enzyme, often covalently; not easily removed by dialysis.	FAD in some flavoproteins, heme in catalase/peroxidase
Activator	Inorganic ion that enhances enzyme activity.	Ca^{2+}, Mg^{2+}, Mn^{2+}, Cl^-

Important relation

Holoenzyme = Apoenzyme + Coenzyme/Cofactor. The apoenzyme mainly determines enzyme specificity, while the coenzyme helps in transfer of atoms or groups such as hydrogen, amino, acyl, methyl, phosphate or carbon dioxide.

5. Proenzymes / Zymogens

Definition

Proenzymes, also called zymogens, are inactive enzyme precursors. They are converted into active enzymes by irreversible cleavage of one or more peptide bonds. This prevents unwanted digestion or damage inside the cells where the enzymes are produced.

Examples of proenzyme activation

Proenzyme	Active enzyme	Activator	Importance
Pepsinogen	**Pepsin**	Gastric HCl and autocatalysis	Protein digestion in stomach
Trypsinogen	**Trypsin**	Enteropeptidase; trypsin also autocatalyses trypsinogen	Activates other pancreatic proteases
Chymotrypsinogen	**Chymotrypsin**	Trypsin	Protein digestion in small intestine
Proelastase	**Elastase**	Trypsin	Digestion of elastin and other proteins
Procarboxypeptidase A/B	**Carboxypeptidase A/B**	Trypsin	Cleaves terminal amino acids from peptides
Plasminogen	**Plasmin**	Tissue plasminogen activator / urokinase	Fibrinolysis; dissolves fibrin clots

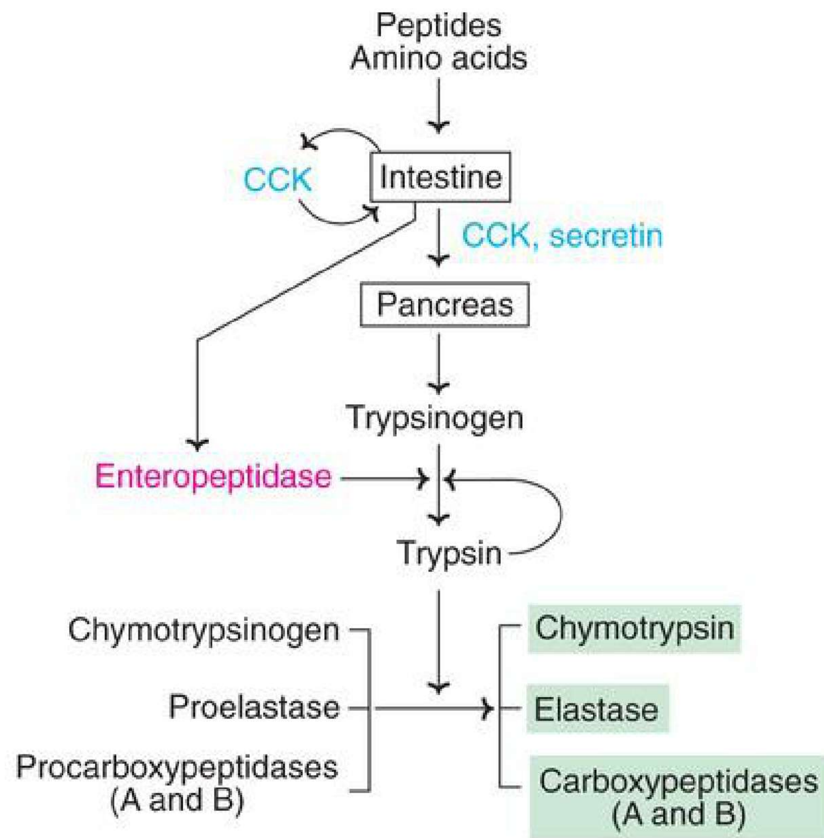

FIG. 8.6 Formation and activation of pancreatic proteases (CCK-Cholecystokinin).

Source Fig. 8.6: Formation and activation of pancreatic protease zymogens.

6. Mechanism of Enzyme Action

Basic equation

$E + S \rightleftharpoons ES \rightarrow E + P$

E = enzyme, S = substrate, ES = enzyme-substrate complex, P = product. The enzyme combines transiently with substrate at the active site, forms ES complex, converts substrate to product, and is released unchanged.

6.1 Enzymes lower activation energy

For any reaction to occur, reactants must reach an activated state called the transition state. Enzymes lower the activation energy needed to reach this state, so reactions occur rapidly at body temperature. Enzymes do not change the equilibrium constant of a reaction; they only increase the rate at which equilibrium is reached.

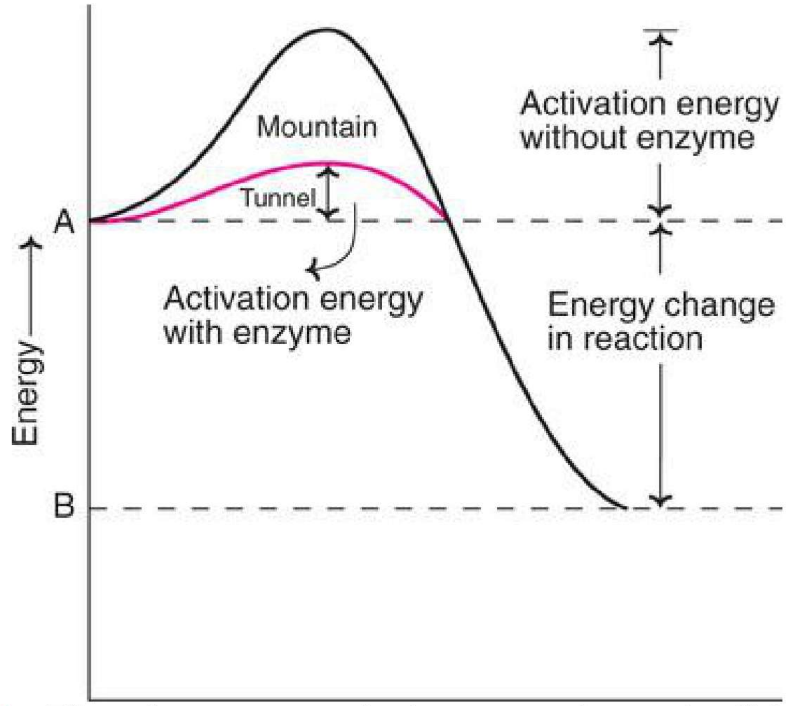

FIG. 6.11 Effect of enzyme on activation energy of a reaction (A is the substrate and B is the product. Enzyme decreases activation energy).

Source Fig. 6.11: Effect of enzyme on activation energy.

Activation-energy concept	Exam meaning
Activation energy	Minimum energy needed to convert substrate into transition state.
Enzyme effect	Provides an easier pathway by lowering the energy barrier.
Equilibrium	Does not change equilibrium constant; only speeds up the reaction.
Body temperature	Allows biochemical reactions to proceed rapidly below 40°C.

6.2 Formation of enzyme-substrate complex

The substrate must bind at the active site before catalysis. Binding positions the substrate correctly, weakens specific bonds and helps form the transition state.

Step	Event	Result
1	Substrate approaches active site	Weak non-covalent binding begins.
2	ES complex forms	Substrate is correctly oriented.
3	Induced fit/strain occurs	Transition state is favored.
4	Catalysis takes place	Bonds are broken or formed.
5	Product is released	Enzyme remains unchanged and reusable.

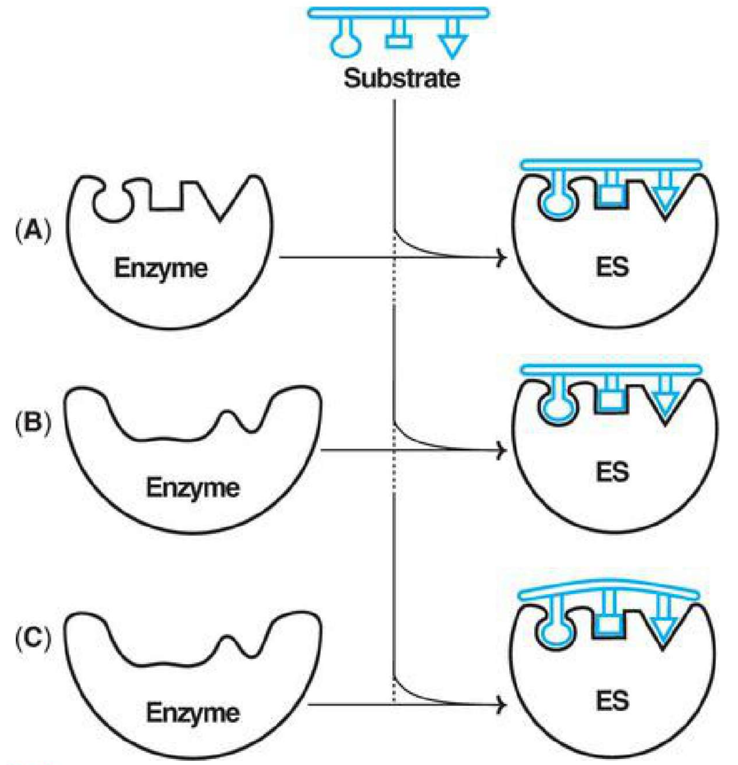

FIG. 6.12 Mechanism of enzymesubstrate (ES) complex formation (**A**) Lock and key model (**B**) Induced fit theory (**C**) Substrate strain theory.

Source Fig. 6.12: Lock-and-key, induced fit and substrate-strain models of ES complex formation.

Models explaining enzyme-substrate binding

Model	Main idea	Importance
Lock and key model (Fischer)	The active site is rigid and pre-shaped. Only the correct substrate fits, like a key in a lock.	Explains specificity but does not explain enzyme flexibility or allosteric effects.
Induced fit model (Koshland)	The active site is flexible. Substrate binding induces a conformational change in enzyme, producing a better fit.	Most accepted model; explains specificity, catalysis and allosteric effects better.
Substrate strain theory	Binding strains the substrate and raises it toward the transition state, making product formation easier.	Often works together with induced fit.

6.3 How enzymes increase reaction rate

- **Acid-base catalysis:** amino acid residues act as proton donors or acceptors. Histidine is important at physiological pH.
- **Covalent catalysis:** enzyme forms a temporary covalent bond with substrate. Example: serine proteases such as trypsin and chymotrypsin.
- **Proximity/orientation effect:** enzyme brings reactants close together and in the correct orientation, greatly increasing reaction rate.
- **Substrate strain:** enzyme binding distorts the substrate, making bonds easier to break and transition state easier to reach.
- **Transition-state stabilization:** enzyme stabilizes the transition state more than the substrate, lowering activation energy.

7. Mechanism / Factors Affecting Enzyme Activity

Enzyme activity means the rate at which an enzyme converts substrate into product. The following factors influence activity:

Factor	Effect on enzyme activity
Enzyme concentration	When substrate is available in excess, reaction velocity is directly proportional to enzyme concentration.
Substrate concentration	Velocity increases with substrate concentration until enzyme active sites become saturated. At saturation, velocity approaches Vmax.
Temperature	Activity increases with temperature up to an optimum; high temperature denatures enzyme and decreases activity.
pH	Each enzyme has an optimum pH. Pepsin works best in acidic pH, while many intracellular enzymes work near neutral pH.
Product concentration	Accumulation of product may slow the reaction or shift equilibrium backward.
Inhibitors	Competitive, non-competitive, irreversible and allosteric inhibitors reduce enzyme activity.
Activators and cofactors	Metal ions and coenzymes can increase activity when required for catalysis.

8. Michaelis Constant (Km) - Quick Revision

- Km is the substrate concentration required to produce half of the maximum velocity (Vmax/2).
- Low Km means high affinity between enzyme and substrate.
- High Km means low affinity between enzyme and substrate.
- Km is a characteristic constant of a given enzyme-substrate pair under fixed conditions.

9. Exam-Oriented Summary

1. Enzymes are protein biocatalysts, mostly synthesized by living cells; ribozymes are RNA exceptions.	2. The six enzyme classes are oxidoreductases, transferases, hydrolases, lyases, isomerases and ligases.
3. The active site is a flexible cleft/pocket where substrate binds and catalysis occurs.	4. Holoenzyme = apoenzyme + cofactor/coenzyme. Coenzymes are often B-vitamin derivatives.
5. Proenzymes or zymogens are inactive precursors activated by cleavage of peptide bonds.	6. Enzyme action involves formation of ES complex, lowering of activation energy and conversion of substrate to product.
7. Induced fit model is more realistic than the rigid lock-and-key model.	8. Enzyme activity is affected by enzyme concentration, substrate concentration, temperature, pH, inhibitors, activators and cofactors.